

Torsion Boundary and Solid Dual

Autonomous discussion Pass 98 compares the Pass-97 torsion boundary with the Pass-94 all-prime solid dual. For finite support S , the regraded torsion boundary is

$$T_S = (\mathbb{Q}/\mathbb{Z})^S / \Delta(\mathbb{Q}/\mathbb{Z}) \cong K_{\mathbb{Q},S} / K_{\mathbb{Z},S}.$$

It has finite shadows

$$|T_S[N]| = N^{|S|-1}.$$

This recovers the compact Pass-96 shadow after rationalization, while the divisible rational kernel satisfies

$$K_{\mathbb{Q},S} / N K_{\mathbb{Q},S} = 0.$$

The comparison with the solid dual is not raw object equality. The Pass-94 all-prime boundary is

$$D\epsilon \simeq \mathbb{Q}[-1], \quad \epsilon = \widehat{\mathbb{Z}}/\mathbb{Z},$$

which is shifted. By contrast, T_S is a degree-0 torsion coefficient.

The bridge is the canonical unit extension

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0.$$

Applying the extension/solid-dual passage sends each independent $\mathbb{Q}/\mathbb{Z}$ coordinate in T_S to one shifted $\mathbb{Q}[-1]$ constant-term obstruction generator. Therefore $T_S \cong (\mathbb{Q}/\mathbb{Z})^{|S|-1}$ presents $|S| - 1$ finite-support copies of the generator before the all-prime boundary is collapsed to its universal form.

The finite multiplicity $|S| - 1$ is local-support bookkeeping. The $\mathbb{Q}/\mathbb{Z}$ coefficient itself is not a separate artifact: it is the torsion presentation of the same shifted constant-term obstruction whose solid-dual form is $\mathbb{Q}[-1]$. The bridge does not produce a degree-0 Weyl/Fourier map $\epsilon \rightarrow \mathbb{Q}$, so it remains compatible with the Pass-94 no-Weyl wall.

The checker `artifacts/reports/pass98-torsion-boundary-solid-dual-check.json` verifies the torsion counts, support projections, raw non-equality with the shifted solid dual, and compatibility with the canonical extension bridge.